National University of Computer and Emerging Sciences, Karachi
FAST School of Computing
Midterm I Examination -- Solution, Fall 2021
15^{th} Oct 2021, 2:30 pm to 3:30 pm

Course Code: CS3001	**Course Name: Computer Networks**
Instructor Names: Dr. Aqsa Aslam, Mr. Shoaib Raza.	
Student Roll No:	**Section No:**

Instructions:

- Return the question paper.
- **All questions must be answered in answer script and according to the sequence given in the question paper.**
- In case of any ambiguity, you may make an assumption, however, your assumption should not contradict any statement in the question paper.

Time: 60 minutes. **Max Points**: 40 points

Question 1: Consider Figure 1, in which a client sends an http request to a server. The path from client to server has three links, of rates R1 = 100 Mbps, R2 = 10 Mbps, and R3 = 1 Mbps, each with link length of 3 km. Suppose that the packet length is L= 18000 bits and speed of light propagation delay on each link is 3*10^8 m/sec.

a) What is the transmission delay of R2?
b) What is the propagation delay of R1?
c) Name the sources of packet delay?
d) What layer in the IP stack moves datagrams from the source host to the destination?

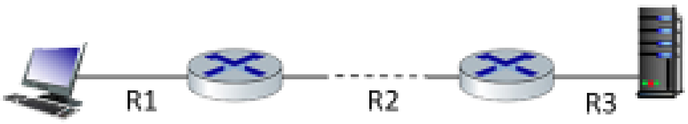


Figure 1

Solution:

a) R2 transmission delay = L/R2= 18000 bits/10 Mbps =0.0018 second
b) R1 propagation delay = d/s= 3 Km / 3*10^8 m/sec = 3*1000/3*10^8 =1.00E-5 sec.
c) Propagation delay, Transmission delay, Queuing delay, processing delay.
d) Network layer

Question 2: Fast.com is registered and hosted with GoDaddy. Both the web server and mail server of Fast are associated with 10.11.21.13 and 10.11.21.14 respectively. The primary authoritative name server of GoDaddy is dns1.GoDaddy.com which is mapped to 172.16.10.1. List down all the resource records (RRs) that will be inserted into the authoritative name server and. com's TLD (top level domain) server.

Solution:

RR Inserted in Authoritative NS
(Fast.com, 10.11.21.13, A)
(Fast.com, 10.11.21.14, MX)

RR Inserted in TLD
(Fast.com, dns1.GoDaddy.com, NS)
(dns1.GoDaddy.com, 172.16.10.1,A)

Question 3: Suppose Huzaifa, with a Web-based e-mail account (such as Yahoo, Hotmail or Gmail), sends a message to Zafeer, who accesses his mail from his mail server using IMAP. Discuss how the message gets from Huzaifa's host to Zafeer's host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts.

Solution: Message is sent from Huzaifa's host to his mail server over HTTP. Huzaifa's mail server then sends the message to Zafeer's mail server over SMTP. Zafeer then transfers the message from his mail server to his host over IMAP.

Question 4: Assume that the client wants to retrieve the www.google.com home page but has no information about the www.google.com web server IP address. When obtaining the IP address for the host name www.google.com, assume the round trip time between local DNS server and DNS root server is 3RTT, between local DNS server and DNS TLD server is 2RTT, and between the clients and the local DNS server is RTT.

a) How long does it take for the first client to obtain the IP address for www.google.com?
b) After that first client, a second client (on the same network) also wants to obtain the IP address for www.google.com. How long for the second client?

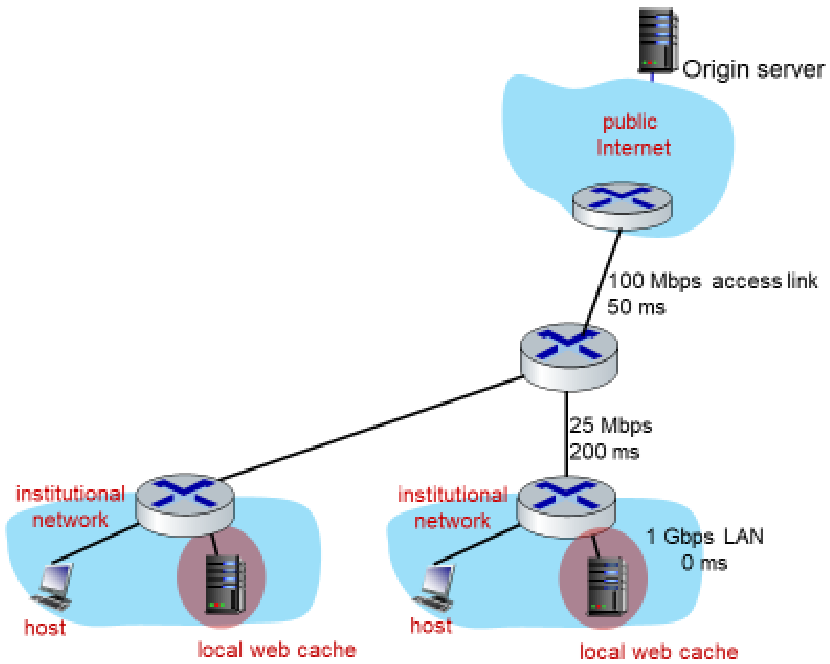
Solution:

a) It takes 6RTT for the first client to obtain the IP address for google.com
b) It takes RTT for the second client (as the IP address is already in local domain server)

Question 5: Consider the scenario is shown in Figure 2, in which origin servers is connected to a router by a 100 Mbps link with a 50ms propagation delay. That router in turn is connected to two routers, each over a 25Mbps link with a 200ms propagation delay. A Gbps link connects a host and a cache (when present) to each of these routers; this link, being a local area network has 0 propagation delay. All packets in the network are 20,000 bits long.

a) What is the end-to-end delay from when a packet is transmitted by the server to when it is received by the client? In this case, we assume there are no caches, there's no queuing delay at the routers, and the packet processing delays at routers and nodes are all 0.

Figure 2



Solution:

If all packets are 20,000 bits long →

it takes 200 usec to send the packet over the 100Mbps link(L/R1)

800 usec to send over the 25Mbps link (L/R2),

20 usec to send over the 1Gbps link(L/R3).

Sum of the three-link transmission is 1020 usec and sum of all propagation is 250ms

Thus, the total end-to-end delay is 251.02 msec.

b) What is the maximum rate at which the server can deliver data to a single client if we assume no other clients are making requests? Here we assume that client hosts send requests for files directly to the server (caches are not used or off in this case).

Solution:

Server send at the max of the bottleneck link 25Mbps.

c) Again we assume only one active client but in this case the caches are on and behave like HTTP caches. A client's HTTP GET is always first directed to its local cache. 60% of the requests can be satisfied by the local cache. What is the average rate at which the client can receive data in this case?

Solution:

We assume that requests are serially satisfied.

40% of the requests can be delivered at 25Mbps and 60% at 1Gbps.

So the average rate is: (0.4×25Mbps+0.6×1Gbps) =610Mbps.

d) What is the average end-to-end delay for the case that 60% of the requests can be satisfied by the local cache?

Solution:

40% of the requests have a delay of 251.02 msec

60% of the requests have a delay of 20 usec.

So the average delay is (100.408ms+12usec) = 100.41 msec.

Best of Luck!